
EVENT-TRADER: AN EVENT-DRIVEN LLM RESEARCH RUNTIME FOR ASYNCHRONOUS FINANCIAL EVIDENCE

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ABSTRACT

LLM trading agents commonly advance through daily or other predefined workflows. At each scheduled decision step, the workflow assembles a research context for LLM analysis. Under asynchronous financial evidence, this design creates a timing trade-off. A longer batch interval makes incoming records wait longer for the next scheduled decision step and can place evidence from different stages of a market episode in one context. A shorter batch interval creates more scheduled decision steps and can turn partial or repeated updates into separate inputs. We present EVENT-TRADER, an event-driven LLM research runtime for continuous financial data streams and controlled replay. It admits each discrete evidence record at its visibility time and publishes a typed message on the bus. The resulting evidence stream can advance the research process as records arrive. Continuous Event-to-Analysis Unit (CEAU) formation controls how related records from this stream become bounded inputs for LLM analysis. Across six months of GCUSD data, EVENT-TRADER achieves a 37.68% return and the strongest risk-adjusted performance among the evaluated agentic and rule-based methods. In the controlled comparison, the event-driven input-formation path records the highest return, Sharpe ratio, and Sortino ratio and the lowest MDD among the five variants. Compared with direct per-record invocation, CEAU reduces total successful analysis runtime by 64.2% and successful token use by 62.9%. The two paths produce similar returns over the one-week ablation. These results show that trading research can follow changing financial evidence while CEAU limits repeated LLM analysis.

Keywords LLM trading agents · event-driven trading · discretionary trading · asynchronous financial evidence · stream processing

1 Introduction

LLM trading agents commonly run as scheduled workflows or loops. At each daily or predefined decision step, the workflow assembles prices, news, memory, and tool outputs for one round of LLM analysis. The agents then update a market view, propose a trading action, and record later feedback Xia et al. [2026]. Recent systems strengthen this round with layered memory, multimodal inputs, tools, role-based analysis, debate, risk review, and reflection Yu et al. [2023], Li et al. [2023], Zhang et al. [2024], Xiao et al. [2024], Yu et al. [2024]. Live and sequential studies keep the available observations current Li et al. [2025], Qian et al. [2025], Chen et al. [2025a], Song et al. [2025]. The research cycle still advances through predefined decision steps.

Market information follows a different clock. We treat it as a data stream: a continuing sequence of discrete records that become visible at uneven times. These records include news reports, filings, official releases, corrections, and source follow-ups. A scheduled workflow waits until the next decision step before using the visible records. A longer batch interval increases this wait and can place the first report, later clarification, and market reaction in one research input. The LLM then receives several stages of the same episode after part of the episode has unfolded. A shorter batch interval forms inputs more often. It can repeatedly analyze partial reports, duplicate coverage, or records that leave the current view unchanged. The batch interval therefore controls both response delay and LLM analysis frequency.

Recent event-centered systems show the value of organizing financial analysis around events. In Janus-Q, event-driven refers to treating financial news events as primary decision units. Its trading evaluation aggregates event signals between successive market opens and executes at the next market open Li et al. [2026a]. FinKario dynamically adds time-sensitive events to a financial knowledge graph and uses the updated graph for weekly prediction Li et al. [2026b]. These systems use events to organize the content used for training, retrieval, prediction, or trading. In this paper, event-driven refers to runtime control over a continuing evidence stream. Each newly visible record can advance the research workflow. Our study examines which records advance the workflow, which related records enter the same input, and when that input becomes ready for analysis.

The runtime distinguishes three objects in the research stream. A *market event* is the development being researched. An *evidence record* is an admitted source record, such as a news report, filing, official release, or web result. A *research artifact* is an attention decision, analysis, portfolio review, reflection result, or memory update produced inside the runtime.

A single market event can unfold through several evidence records. Event studies measure price behavior around discrete information releases MacKinlay [1997]. Consider a central-bank decision. The decision is the market event. The policy statement, press-conference remarks, transcript, and later reports are separate evidence records. The observed price path is part of the visible market state, which the runtime reads separately at the decision time. An attention decision, analysis, or thesis update is a research artifact. These evidence records become visible at different times. We use *asynchronous financial evidence* for this arrival pattern. When a research operation becomes eligible, the runtime forms its input from the evidence records and prior research artifacts visible at that operation’s decision time. It also reads the visible market state required by the operation.

EVENT-TRADER implements this control rule with a message bus. In live operation, market-news WebSocket and web-source adapters produce discrete source inputs as they arrive. Admission stores each accepted input as a time-visible evidence record and publishes a typed message that refers to it. The continuing sequence of visible records forms the evidence stream consumed by the research runtime. A newly visible record can advance the next eligible operation without waiting for a fixed calendar boundary. Market-data adapters maintain the visible market state used by these operations. In controlled replay, archived records enter at their recorded availability times and reconstruct the same stream. Live operation and replay therefore use the same record types, visibility rules, and research path.

An event-driven research loop still needs a rule for expensive LLM analysis. Direct per-record invocation turns every evidence arrival into a separate analysis unit, so analysis volume follows stream volume. One market event can produce an initial report, repeated coverage, and later confirmation while the market reaction continues to develop. Continuous Event-to-Analysis Unit (CEAU) formation controls how this evidence stream becomes analysis units. It lets related records wait, accumulate, or form a bounded unit. Material evidence can proceed promptly. A developing episode can remain open as related evidence arrives and the visible market state advances. CEAU therefore preserves event-driven processing while controlling when accumulated evidence enters one LLM analysis.

These choices lead to two system contributions and one empirical study:

- **An event-driven LLM research runtime.** We present EVENT-TRADER, which stores a continuous market stream as time-visible evidence records and coordinates research through typed bus messages that refer to durable records and artifacts. These messages advance attention review, analysis, portfolio review, reflection, and memory operations. The runtime separates source evidence, revisable research memory, executed portfolio state, and delayed feedback.
- **Continuous Event-to-Analysis Units.** We introduce CEAU, which groups related evidence into bounded analysis units and controls when each unit enters LLM analysis. Each emitted unit fixes its source evidence and the analysis boundary used to assemble visible context for one analysis.
- **An evaluation aligned with the two design steps.** We compare EVENT-TRADER with three agentic baselines and five rule-based strategies over six months of GCUSD data. EVENT-TRADER returns 37.68%. We then evaluate input formation through two controlled comparisons. The first compares the event-driven input-formation path, including CEAU, with four scheduled settings. The second compares CEAU with direct per-record invocation within the event-driven path. The event-driven path records the highest return, Sharpe ratio, and Sortino ratio and the lowest MDD among the five input-formation variants. CEAU reduces total successful analysis runtime by 64.2% and successful token use by 62.9%. Over the one-week ablation, CEAU and direct per-record invocation return 1.53% and 1.44%, respectively.

2 Background

Discretionary trading. A research view records the current interpretation of a target. It may contain a thesis, catalysts, risks, conviction, and conditions for review. A *thesis* explains why the target may reprice. A *catalyst* is a development that may change this explanation or accelerate repricing. *Conviction* records the strength of the current view.

Discretionary trading revises this view through contextual judgment under uncertainty. Information acquisition is costly Grossman and Stiglitz [1980]. Limited attention shapes the set of stocks considered by individual investors Barber and Odean [2008]. Studies of institutional portfolios find asymmetric patterns across buying and selling decisions and relate them to attention allocation Akepaniditaworn et al. [2023]. Portfolio state records executed exposure. A portfolio review maps the current research view and risk constraints to target exposure.

Large language models. An LLM is a Transformer-based generative model that maps a token sequence to a probability distribution over following tokens Vaswani et al. [2017]. The *context window* is the token sequence available to one model call. Instructions, retrieved documents, tool results, memory, and prior messages can all enter this window. Financial language models extend general language modeling with financial corpora, instruction data, or domain benchmarks Wu et al. [2023a], Yang et al. [2023], Xie et al. [2023, 2024].

LLM financial research agents. An LLM agent combines a language model with state, memory, tools, and a control loop Wang et al. [2023]. At each step, the agent receives an observation, selects a tool call or structured output, receives the result, and updates its working context. ReAct interleaves reasoning with actions over external tools Yao et al. [2023]. Reflexion stores verbal feedback in episodic memory for later trials Shinn et al. [2023]. Multi-agent systems assign different roles to several agents and coordinate their messages or artifacts Wu et al. [2023b].

LLM financial research agents apply these capabilities to research-view maintenance and portfolio review. Representative systems use persistent memory Yu et al. [2023], Li et al. [2025], tool-based access to financial information Xiao et al. [2024], Tian et al. [2025], Li et al. [2025], and role-specific analysis Xiao et al. [2024], Tian et al. [2025], Chen et al. [2025b], Li et al. [2025].

A research operation reads available evidence records, visible market state, and eligible prior research artifacts. Analysis may revise the research view. Portfolio review combines research material with current exposure and risk constraints. Here, *event-driven* describes the runtime rule under which a newly visible evidence record can make a research operation eligible. The trading horizon remains discretionary.

3 Existing Paradigms Revisited

Existing financial systems differ in their data sources, tasks, and agent roles. However, most of the work reviewed here still processes accumulated information in batches at predefined decision times, when LLM-based analysis or trading decisions are triggered. We examine this fixed-time design across two lines of work.

Workflow-style LLM trading agents. Workflow-style LLM trading agents run research and trading as a repeated sequence Xia et al. [2026]. At each decision time, the system forms a research input from market data, news, memory, and portfolio state. It then runs analysis, review, and trading operations. Existing systems add different forms of memory, tools, agent roles, debate, feedback, and risk control Yu et al. [2023], Zhang et al. [2024], Xiao et al. [2024], Yu et al. [2024], Li et al. [2025], Song et al. [2025]. The exact workflow and schedule differ across systems. StockBench forms daily market signals Chen et al. [2025a]. LiveTradeBench forms an input at each trading step Yu et al. [2025]. In both cases, the decision time determines when current market information enters the workflow.

Event-centered financial systems. Event-centered systems use market events to structure financial information and form prediction or trading inputs. Janus-Q treats financial news events as decision units and associates each event with its subsequent market response Li et al. [2026a]. Its trading evaluation aggregates these signals and executes at the next daily market open. FinKario updates a financial knowledge graph as company information and time-sensitive events arrive Li et al. [2026b]. The updated graph supports prediction on a weekly schedule. In both systems, events organize the information used by the model, while prediction or trading still occurs at predefined times.

Time-batched workflows. The distinction lies in what advances the downstream workflow. Evidence may be organized around events or used to update a knowledge base. In the evaluated prediction and trading workflows, however, its arrival does not directly trigger downstream decision processing. The workflow waits until the next predefined decision time, groups the records collected since the previous decision time, and forms them into a research

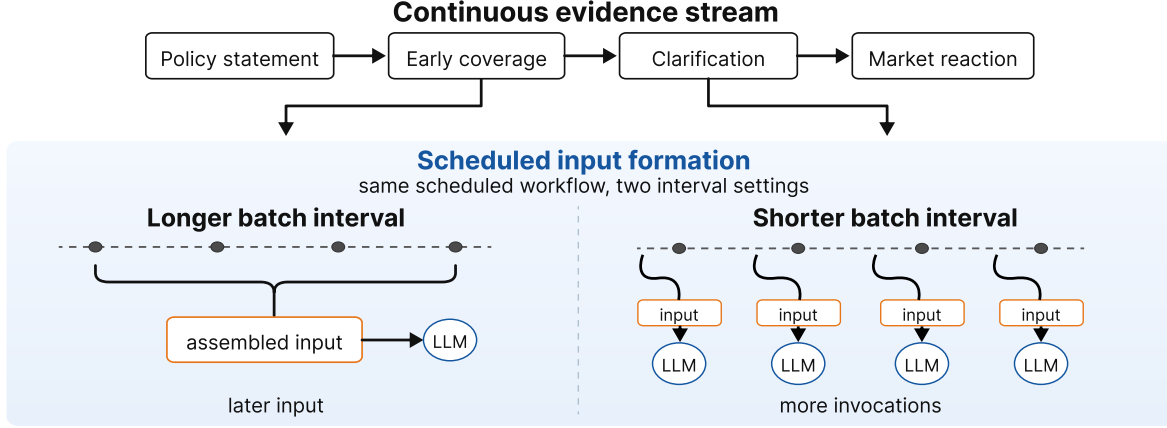


Figure 1: Time-batched input formation from the same continuous evidence stream under longer and shorter fixed intervals.

input. We refer to this pattern as a *time-batched workflow*. Figure 1 illustrates how the same evidence stream is converted into research inputs under longer and shorter fixed intervals.

A longer interval delays the use of newly arrived evidence. It may also combine several stages of an evolving market event into one research input. By the time the LLM receives this input, the event may already have developed substantially. A shorter interval reduces this delay but invokes the workflow more frequently. The system may repeatedly analyze an event before sufficient evidence has accumulated. It may also run again when a newly arrived record does not materially change the current research state. A single batch interval therefore couples response latency with analysis frequency.

4 Method

4.1 EVENT-TRADER Framework

EVENT-TRADER implements event-driven input formation by processing admitted evidence as a continuous data stream. Live and replay modes use the same research path. In live mode, market-news and web-source adapters publish source inputs as they become available. Market-data adapters maintain the visible market state used by analysis and maturity checks. In replay mode, archived source inputs are released at their recorded visibility times to reconstruct the same evidence stream.

The runtime separates durable records from the messages that coordinate research operations. Admission writes accepted source records to the evidence ledger. The message bus carries typed messages with event time, recorded time, target, causal links, and references to durable records. Downstream operations consume the evidence stream through these messages under the same visibility boundary. A newly visible evidence record can therefore advance the research path without waiting for a scheduled decision step.

Figure 2 shows this path. Source adapters feed evidence admission and the evidence ledger. The Checker Agent decides whether an admitted evidence record deserves analysis attention. Continuous Event-to-Analysis Unit (CEAU) formation controls when related evidence becomes a bounded analysis unit. The Analysis Agent consumes emitted units and writes research artifacts. The PM Review Agent combines analysis with portfolio state. Delayed reflection can make selected memory artifacts available to later research.

Evidence acquisition and classification. Each admitted evidence record carries one `event_type` label and one `source_kind` label from fixed vocabularies. Event type describes the subject of the record, such as monetary policy, macro data, geopolitical security, or price action. Source kind describes the form of the material, such as an official release, news report, data page, commentary, or research note. A deterministic validator checks the label count and vocabulary.

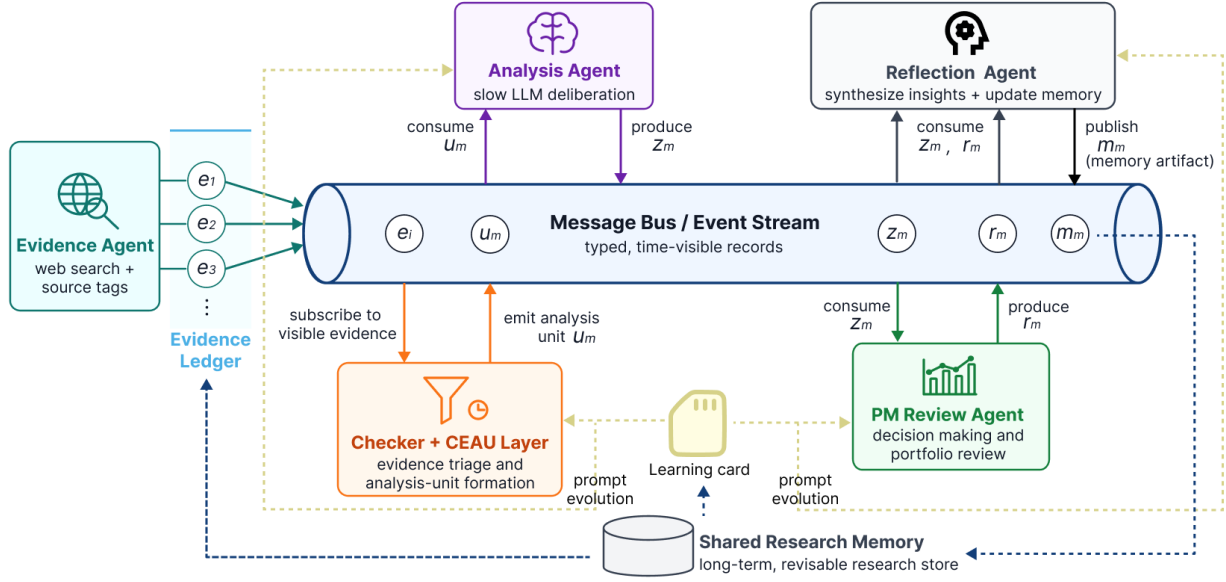


Figure 2: The event-driven research runtime of EVENT-TRADER.

These labels organize the evidence before attention review. Evidence-review fields add an evidence role and a price-recap marker for later routing and grounding. Importance, urgency, thesis impact, and trading direction remain downstream judgments. The classification fields therefore describe the record and support routing. Portfolio action is decided later.

Event-stream coordination. The message bus coordinates the runtime through typed messages. An admission message refers to an evidence record in the evidence ledger. A Checker decision refers to both its source evidence and its recorded attention decision. An emitted-unit message refers to the bounded input prepared by CEAU. Analysis, portfolio review, and reflection publish their own completion messages and durable artifacts.

Durable state remains outside the message bus. The evidence ledger keeps append-only source records. Research memory keeps revisable analysis and claims. Portfolio state records executed exposure and changes after execution. Message references preserve the path between these stores and the operation that produced or consumed each record.

Each operation consumes a defined message type. An admitted evidence record makes attention review eligible. A Checker decision advances evidence to CEAU routing when further research is required. An emitted analysis unit makes LLM analysis eligible. A completed analysis can make portfolio review eligible and can create a later reflection obligation. The message and its recorded time determine when each handoff can occur.

Evidence triage and analysis-unit formation. The Checker Agent receives one admitted evidence record and a bounded context pack. The pack contains the active evidence and eligible research state visible at the decision time. The Checker records whether the evidence deserves analysis attention. A conservative validator sends uncertain or unsupported quiet decisions forward for analysis. Thesis formation, memory revision, portfolio review, and trading remain downstream.

Evidence that proceeds beyond the attention gate enters CEAU formation. Event-driven coordination determines when records become available to research. Analysis frequency requires a separate control. Direct per-record invocation gives every admitted record its own slow LLM analysis. CEAU instead allows related evidence to wait, accumulate, or become a bounded analysis unit. Section 4.2 defines the formation rules.

Grounded analysis and portfolio review. The Analysis Agent consumes an emitted analysis unit through a ReAct-style tool-use process. A context packet fixes the unit’s primary evidence, eligible memory reads, visible market context, runtime scope, and decision time. Grounding checks link the active evidence to the compiled context or later tool

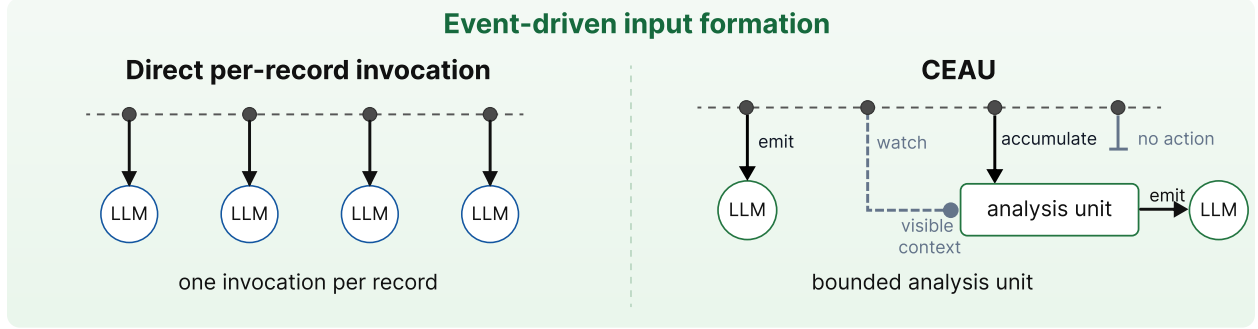


Figure 3: Direct per-record invocation and CEAU-controlled formation from the same admitted evidence stream.

reads. Price-recap evidence also requires visible market data. The analysis outcome retains the context-packet identifier, packet hash, and grounding record.

Analysis is exposure-blind. It can update a claim, revise research memory, record a risk, or prepare material for portfolio review. The PM Review Agent is exposure-aware. It consumes the analysis artifact together with current exposure, review history, visible market state, and PM-visible memory. Execution records the selected market bar, transaction cost, and resulting position state.

Delayed feedback and self-evolution. Outcome-dependent feedback enters the research loop after its evaluation horizon becomes observable. Reflection can write an episode-memory update or an explicit no-update record. A finalized update may later become a role-scoped learning card. Each card has a `usable_from` time. Later analysis or portfolio review can read the card after that time.

4.2 Continuous Event-to-Analysis Unit Formation

Once an evidence record has been admitted and its corresponding Checker decision becomes available, it can enter LLM analysis. Figure 3 contrasts two ways of forming analysis inputs from the same admitted evidence stream. Direct per-record invocation creates a separate analysis unit for every record. This minimizes handoff latency, but makes the number of LLM invocations grow with stream volume and may repeatedly analyze an evolving event before sufficient related evidence has accumulated. Continuous Event-to-Analysis Unit (CEAU) formation instead processes Checker decisions in visibility order and determines whether each record should trigger immediate analysis, enter a pending unit with related evidence, remain available for later matching, or end at the current routing boundary. In this way, CEAU preserves record-level response to the evidence stream while controlling how records are converted into LLM analysis units.

Routing and formation scope. The routing policy combines the Checker decision with deterministic controls and assigns one of four routes. The `emit` route creates an immediate single-record analysis unit. The `accumulate` route adds the record to a compatible pending unit. The `watch` route records a stable watch identifier so that unresolved evidence can be matched with later arrivals. The `no_action` route ends CEAU processing for the current record. Each routing decision produces a durable receipt containing the final route, its reason codes, the associated Checker decision, and the routing-policy version.

Configured interrupt lanes and selected event or source types can override the default lane policy and force `emit`. Other records follow the route permitted by their policy lane. An absence- or maturity-based `no_action` decision is accepted only when it is supported by explicit evidence or when the required source stream has advanced far enough to satisfy its completeness condition. Otherwise, the router changes the decision to `watch`, keeping the unresolved record available for later matching.

For records routed to `accumulate`, the routing policy derives a formation scope that acts as a compatibility key. Only records assigned to the same scope may enter the same pending analysis unit. The event family combines the record’s event type, source kind, and evidence role. Under the default policy, the scope consists of the research target, routing lane, event family, event-time hour, research-memory scope, and market-data scope. Under the context-window policy, it consists of the research target, research-memory scope, and market-data scope. The event-time-hour field limits

which records may enter the same unit; records are still routed when their Checker decisions become visible rather than at hourly invocation boundaries.

Bounded pending units. A pending analysis unit retains its formation scope, primary- and supporting-evidence identifiers, opening time, latest event time, event-time deadline, accumulated character count, and policy version. Primary identifiers refer to records accumulated into the unit. Supporting identifiers preserve explicit links to evidence used as contextual support. These identifiers maintain source-level provenance even when several records are analyzed together.

The event-time deadline of a pending unit is

$$t_{\text{deadline}} = t_{\text{event},1} + N_{\text{wait}}\Delta_{\text{bar}},$$

where $t_{\text{event},1}$ is the event time of the first accumulated record, N_{wait} is the maximum number of bars assigned by the routing policy, and Δ_{bar} is the duration of one market bar. The deadline is fixed when the unit opens. Later records update the latest event time, evidence identifiers, and accumulated content without extending the original waiting boundary. This prevents continued arrivals from postponing analysis indefinitely.

Each routing policy also defines a maximum aggregate character count. If adding an arriving record would exceed this limit, CEAU emits the current pending unit before opening or updating a unit with the new record. The limit controls aggregation rather than truncation: each source record remains atomic and is never split across units. A *bounded analysis unit* is therefore governed by both a maximum waiting horizon and a maximum aggregation size.

Emission and completeness. An analysis unit may be emitted through five conditions. The `emit` route creates an immediate single-record unit. Event-time progress closes a pending unit when its deadline and associated completeness conditions are reached. The aggregation limit closes the current unit before another record is added. Controlled replay drains all remaining units at the end of the replay window. In live execution, a processing-time safety close prevents a unit from remaining open indefinitely when event-time progress stalls.

A watermark records how far a source domain has advanced in event time Akidau et al. [2015]. A completeness requirement pairs a source domain with an event-time boundary that the domain must reach. For example, a market-reaction requirement is satisfied when the market-bar watermark reaches the pending unit’s deadline. Each emitted unit records its satisfied and missing completeness requirements together with its emission reason and emission clock. A processing-time safety close explicitly records incomplete event-time coverage rather than treating the unit as event-time complete.

Algorithm 1 summarizes the record-driven routing and formation procedure. Deadline, replay-drain, and processing-time closures use the same emission operation when their corresponding boundaries are reached.

An emitted analysis unit fixes its source evidence, formation scope, timing fields, emission reason, completeness status, and routing-policy version. The Analysis Agent then constructs a context packet by adding eligible research-memory entries and market state visible at the unit’s analysis boundary. Source evidence, CEAU formation state, and subsequently generated research artifacts therefore remain separately identifiable and auditable.

5 Evaluation

The evaluation addresses three questions. First, how does the complete EVENT-TRADER runtime perform against agentic and rule-based trading methods? Second, what changes when the same downstream analysis and execution process receives event-driven inputs instead of inputs formed at fixed time intervals? Third, how does Continuous Event-to-Analysis Unit (CEAU) formation affect repeated analysis, cost, and trading behavior relative to direct per-record invocation? The first comparison evaluates complete systems under their native workflows. The second holds downstream analysis and execution fixed and compares the event-driven input-formation path, including CEAU, with four fixed-time settings. The third keeps the remaining event-driven path unchanged and replaces CEAU with direct per-record invocation.

5.1 Experimental Setup

Market and replay setting. The study uses gold futures as a single-instrument discretionary trading setting. The overall comparison covers January 1 through June 30, 2026. Market data come from a canonical GCUSD price ledger. Five-minute bars support replay, execution marks, and market-reaction maturity checks. All agentic methods use

Algorithm 1 CEAU routing and analysis-unit formation

Require: Checker decisions in visibility order, evidence ledger, routing policy, durable pending-unit map, source watermarks

Ensure: Routing receipts and emitted analysis units

```

1: for all decision in Checker decisions do
2:   record  $\leftarrow$  evidence record referenced by decision
3:   watermark  $\leftarrow$  latest source-watermark state
4:   emit pending units whose event-time closure conditions are satisfied
5:   (route, receipt)  $\leftarrow$  apply routing policy to decision, record, and watermark
6:   persist receipt
7:   if route = emit then
8:     scope  $\leftarrow$  formation scope for record
9:     unit  $\leftarrow$  create single-record analysis unit
10:    persist and publish unit
11:   else if route = accumulate then
12:     scope  $\leftarrow$  formation scope for record
13:     if a pending unit exists for scope then
14:       if adding record would exceed the aggregation limit then
15:         emit the pending unit with reason context_budget
16:         remove the emitted unit from the pending-unit map
17:       end if
18:     end if
19:     open or update the pending unit for scope
20:   else if route = watch then
21:     retain the watch identifier stored in receipt
22:   else if route = no_action then
23:     finish CEAU processing for the current record
24:   end if
25: end for

```

MiniMax M2.1. The model was released on December 23, 2025, so the complete evaluation window follows its release. This follows DeepFund’s concern about temporal overlap between model training and evaluation data Li et al. [2025], MiniMax [2025]. The results cover one instrument and a controlled replay.

Visibility boundary. At each decision time, a method may use market bars, evidence records, research artifacts, memory records, and feedback artifacts whose visibility time is no later than that decision time. This boundary applies to every evaluated method. External baselines receive only the inputs available at their native decision boundary.

Temporal records. Each EVENT-TRADER analysis stores its decision time, context-packet identifier and hash, active evidence, research-memory reads, and market context. Analysis outcomes, portfolio reviews, execution records, and reflection artifacts carry their own availability times. These records allow us to inspect the information available to each operation.

Execution rules. All portfolio evaluations start with \$100,000 and charge 0.5 basis points per side on executed notional. External baselines retain their native decision schedules, action spaces, position management, and execution rules. In the input-formation and CEAU studies, each directional assessment defines one target exposure. We map *strong_long*, *weak_long*, *flat*, *weak_short*, and *strong_short* to 1.0, 0.5, 0.0, -0.5 , and -1.0 . A new exposure takes effect at the next available five-minute GCUSD bar and remains active until another assessment changes it. The final exposure is marked to market at the end of the evaluation window.

Trading metrics. We report total return, Sharpe ratio, Sortino ratio, maximum drawdown (MDD), executed actions, and transaction fees. Total return measures the change in portfolio value over the reported window. Sharpe and Sortino ratios use a common annualization factor of 310.552 observations per year, derived from the shared GCUSD evaluation calendar. MDD is the largest peak-to-trough loss. One executed action is an opening, closing, addition, reduction, or material rebalance. A reversal counts as two actions. Fees are charged on the resulting executed notional.

Table 1: Overall comparison on GCUSD from January 1 to June 30, 2026. Bold and underlined values denote the best and second-best performance. MDD is the drawdown magnitude, where lower is better. Fees are charged at 0.5 bps per side on executed notional.

Method	Return $\uparrow$	Sharpe $\uparrow$	Sortino $\uparrow$	MDD $\downarrow$	Actions	Fees (\$)
<i>Agentic methods</i>						
EVENT-TRADER (ours)	37.68%	2.878	3.470	10.34%	118	377.77
TradingAgents	18.97%	1.233	<u>1.955</u>	<u>11.64%</u>	30	115.18
FinMem	11.72%	0.918	0.852	12.28%	58	329.31
AI Hedge Fund	2.96%	0.351	0.407	19.74%	127	544.44
<i>Rule-based strategies</i>						
TSMOM (20-day)	20.46%	1.270	1.668	15.82%	104	92.94
MA (10EMA/50SMA)	<u>22.02%</u>	<u>1.398</u>	1.733	13.11%	95	49.15
MACD (12,26,9)	-14.79%	-0.750	-0.956	27.17%	125	127.65
RSI (14)	-16.13%	-0.911	-1.430	17.85%	56	22.11
Bollinger Bands (20,2)	-6.94%	-0.419	-0.454	13.43%	30	45.47
<i>Passive reference</i>						
Buy-and-Hold	-6.96%	-0.264	-0.307	25.30%	2	9.65

Analysis metrics. An analysis trigger is one invocation created by an input-formation rule. A successful analysis completes with a valid analysis artifact and enters the runtime and token aggregates. A directional assessment is a successful analysis that produces one of the five directional states. A repeated state matches the preceding directional assessment after the initial flat reset. A state change differs from the preceding assessment. Runtime and token totals are computed over successful analyses unless a table states that it reports native baseline runs.

5.2 Trading Baselines

We evaluate two groups of active baselines and one passive reference. The first group contains agentic trading systems. TradingAgents uses analyst roles, debate, risk review, and portfolio-manager synthesis Xiao et al. [2024]. FinMem uses layered memory to support daily trading decisions Yu et al. [2023]. AI Hedge Fund uses analyst modules followed by risk management and portfolio-facing synthesis virattt [2026].

The second group contains five rule-based strategies. TSMOM uses the direction of the preceding 20-day return. MA compares a 10-day exponential moving average with a 50-day simple moving average. MACD uses the standard 12, 26, and 9-day configuration. RSI uses a 14-day window. Bollinger Bands use a 20-day window and two standard deviations. Buy-and-Hold serves as a passive market reference.

All evaluated methods use the GCUSD price ledger, evaluation window, visibility boundary, initial capital, and transaction-cost rate defined in Section 5.1. The agentic methods use the same LLM backbone. Each baseline otherwise retains its native decision schedule, workflow, action space, position management, and execution semantics.

TradingAgents uses its final portfolio-manager rating. Buy and Overweight set long exposure, Hold maintains the current exposure, and Underweight and Sell set short exposure. FinMem uses its repository action score, where buy, hold, and sell map to +1, 0, and -1 over the current-close to next-close interval. AI Hedge Fund retains its native buy, sell, short, cover, and hold decisions. Its post-trade state maps to full long, flat, or full short exposure. TradingAgents and AI Hedge Fund execute position changes at the next available five-minute GCUSD bar. The common transaction-cost rate applies to each evaluated exposure change.

5.3 Overall Comparison with Trading Baselines

The first experiment compares complete trading systems over the six-month window. The three agent baselines use daily research workflows with different forms of role coordination, memory, and portfolio synthesis. The rule-based strategies provide active timing references without LLM analysis. Each method retains its native workflow and action semantics. This experiment therefore evaluates end-to-end outcomes. The next experiment holds downstream analysis and execution fixed and compares the event-driven input-formation path, including CEAU, with fixed-time input formation. Figure 4 shows the cumulative returns of the agent systems. Table 1 reports the full comparison.

Trading performance. Figure 4 and Table 1 show that EVENT-TRADER records a return of 37.68%. This is 18.71 percentage points above TradingAgents, the highest-return external agent. EVENT-TRADER also has the highest Sharpe and Sortino ratios and the lowest MDD in the table.

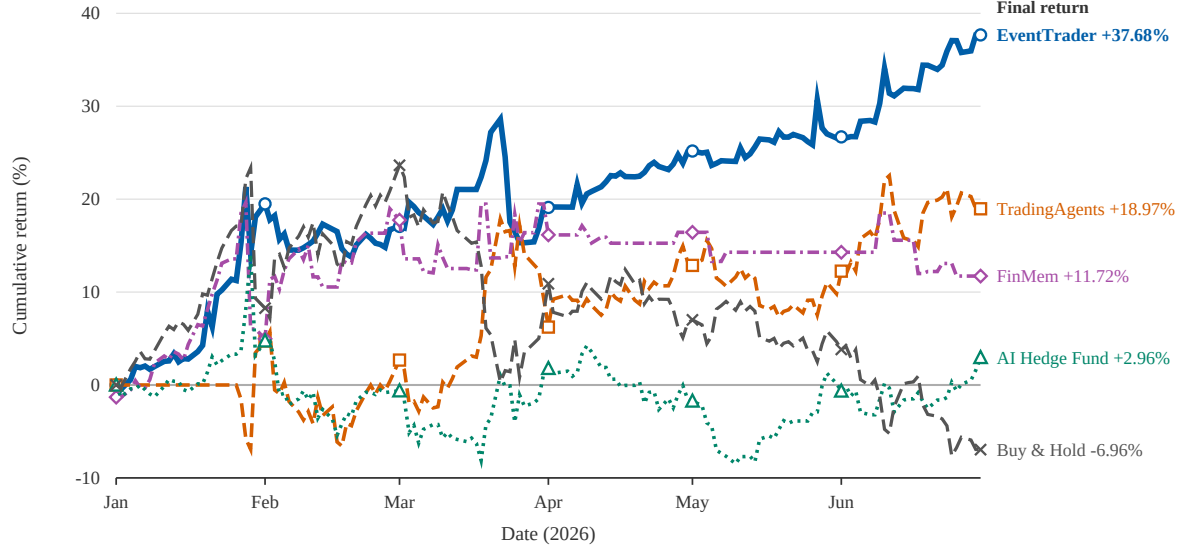


Figure 4: Cumulative GCUSD returns of EVENT-TRADER, the three external agent systems, and Buy-and-Hold from January 1 to June 30, 2026. Each system retains its native decision and execution semantics under the shared market window and visibility boundary.

Table 2: Analysis cost of the agentic methods from January 1 to June 30, 2026. Runs are completed native analyses or decisions. Calls include all model turns within each run. Runtime is observed for all methods.

Method	Runs	Calls/run	Avg. time (s)	Total time (h)	Tokens/run (K)	Tokens (M)
EVENT-TRADER (ReAct)	879	12.7	138.1	33.71	267.3	234.97
TradingAgents	154	16.0	333.0	14.24	121.1	18.64
FinMem	154	1.4	30.2	1.29	5.4	0.83
AI Hedge Fund	129	6.0	46.0	1.64	3.6	0.47

MA and TSMOM also produce positive returns, while Buy-and-Hold loses 6.96%. These results show that the window contains opportunities for active timing. EVENT-TRADER exceeds MA, the highest-return rule-based strategy, by 15.66 percentage points and records a lower MDD, at 10.34% compared with 13.11%. The methods differ in workflow, decision schedule, and action semantics. Attribution to input formation therefore comes from the controlled comparison in Section 5.4.

Analysis cost. Table 2 shows the cost of event-level research coverage. EVENT-TRADER completes 879 analyses, compared with 154 decisions for TradingAgents and FinMem and 129 for AI Hedge Fund. Its mean analysis time is 138.1 seconds, below the 333.0 seconds of TradingAgents. The larger number of analyses raises its total runtime to 33.71 hours. Aggregate runtime therefore follows analysis frequency even when individual analyses are not the slowest.

Token use reflects both event coverage and analysis depth. At the observed mean tokens per call, one call for each of the 879 analysis units would use an estimated 18.5 million tokens. The measured ReAct path uses 12.7 calls per unit and 234.97 million tokens. The observed token cost therefore motivates control over when newly visible evidence becomes another slow analysis.

The next experiment holds the downstream analysis and execution process fixed and varies how the analysis input is formed.

5.4 Event-Driven versus Time-Batched Input Formation

This controlled comparison asks how input formation affects the trading outcome when downstream analysis and execution are held fixed. Every variant receives the same evidence stream and uses the same model backbone, analysis-assessment schema, GCUSD price ledger, exposure mapping, and execution rule. The event-driven variant forms analysis units through CEAU as records arrive. Each time-batched variant evaluates fixed UTC boundaries and forms at

Table 3: Event-driven and time-batched input formation on GCUSD from February 4 to March 5, 2026. Bold values denote the best performance. MDD is the drawdown magnitude, where lower is better. Fees are charged at 0.5 bps per side on executed notional.

Variant	Return $\uparrow$	Sharpe $\uparrow$	Sortino $\uparrow$	MDD $\downarrow$	Assessments	Actions	Fees (\$)
Event-driven	0.47%	0.423	0.684	3.58%	123	31	81.17
Time-batch (5 min)	-5.24%	-3.094	-2.695	6.22%	144	25	61.06
Time-batch (1 h)	-0.22%	-0.021	-0.021	7.37%	127	25	65.40
Time-batch (4 h)	-4.71%	-2.416	-2.064	4.94%	90	18	48.79
Time-batch (1 day)	-5.70%	-3.982	-3.686	5.70%	26	15	36.66

Table 4: Analysis timing and token use from February 4 to March 5, 2026. Runtime and token values are computed over counted successful analyses. The values are descriptive rather than ranked because the variants trigger different numbers of analyses.

Variant	Analysis triggers	Avg. runtime (s)	Total runtime (h)	Tokens (M)
Event-driven	166	155.9	7.10	44.14
Time-batch (5 min)	197	144.0	7.68	47.54
Time-batch (1 h)	171	128.9	6.02	37.80
Time-batch (4 h)	120	110.7	3.69	26.04
Time-batch (1 day)	30	161.9	1.35	7.41

most one input per target at each boundary. An input is formed only when at least one new evidence record has become visible after the preceding boundary and no later than the current boundary. Empty intervals do not trigger analysis. We evaluate boundaries of five minutes, one hour, four hours, and one day. The comparison covers February 4 through March 5, 2026.

Trading performance. Table 3 shows that event-driven input formation returned 0.47% and was the only positive variant in this window. The one-hour schedule had the highest return among the fixed schedules, at -0.22% . Event-driven input formation also had the highest Sharpe and Sortino ratios and the lowest MDD.

The number of directional assessments decreased from 144 at five-minute boundaries to 26 at daily boundaries. Event-driven formation produced 123 assessments, close to the 127 assessments of the one-hour schedule. The two variants ended with returns of 0.47% and -0.22% , respectively. Across all four fixed schedules, assessment counts ranged from 26 to 144 and every return was negative. Assessment frequency therefore overlaps across the settings, while their trading outcomes differ in this window.

Analysis frequency and cost. Table 4 reports both analysis frequency and aggregate cost. The five-minute schedule produced 197 triggers, 7.68 hours of counted analysis runtime, and 47.54 million tokens. Event-driven formation produced 166 triggers, 7.10 hours, and 44.14 million tokens. It therefore used less aggregate analysis cost than the five-minute schedule.

Longer fixed intervals reduced both trigger counts and aggregate cost. The one-hour schedule used 6.02 hours and 37.80 million tokens. The four-hour and daily schedules reduced these totals further. Their returns were -0.22% , -4.71% , and -5.70% , respectively. Event-driven formation used more analysis cost than these three schedules and was the only variant with a positive return. These results describe the cost and trading outcomes observed under the evaluated input-formation rules. The next experiment compares CEAU with direct per-record invocation within the event-driven path.

5.5 CEAU Ablation

Direct per-record invocation treats every admitted evidence record as a separate analysis input. CEAU keeps related records in a pending unit and may emit them together. We compare these two paths from February 24 to March 2, 2026. Both paths use the same evidence source, model backbone, analysis schema, GCUSD price ledger, and execution rule. The experiment measures analysis frequency, repeated directional assessments, cost, and trading outcomes.

Repeated analysis and cost. Table 5 shows how the two paths turn arriving records into LLM analyses. Direct per-record invocation creates 78 analysis triggers from 78 visible records. It produces 60 directional assessments, of

Table 5: CEAU formation ablation on GCUSD from February 24 to March 2, 2026. MDD is reported as a positive magnitude. Runtime reports mean seconds per successful analysis and total hours over the same calls. Repeated states count directional assessments that match the preceding state after the initial flat reset. State changes are counted over the timestamp-ordered assessment sequence. Bold values mark the preferred direction for return, MDD, fees, triggers, repeated states, runtime, and token use. Actions, assessments, and state changes are descriptive.

<i>Trading outcome and execution</i>						
Path	Return $\uparrow$	MDD $\downarrow$	Actions	Fees (\$) $\downarrow$		
Direct per-record invocation	1.44%	0.27%	3	15.03		
CEAU formation path	1.53%	0.48%	2	5.01		

<i>Analysis behavior and cost</i>						
Path	Triggers $\downarrow$	Assessments	Repeated states $\downarrow$	State changes	Runtime (s / h) $\downarrow$	Tokens (M) $\downarrow$
Direct per-record invocation	78	60	55	5	133.7 / 2.90	19.44
CEAU formation path	33	21	19	2	120.5 / 1.04	7.22

which 55 repeat the preceding state. CEAU produces 33 triggers and 21 directional assessments. Nineteen of these assessments repeat the preceding state.

Repeated states account for 91.7% of the direct per-record assessments and 90.5% of the CEAU assessments. Repeated assessments therefore remain common in both paths. The difference lies in their total number. Direct per-record invocation produces 39 additional directional assessments, including 36 additional repeated states.

Mean runtime per successful analysis is 133.7 seconds for direct per-record invocation and 120.5 seconds for CEAU. Across the full window, total successful analysis runtime falls from 2.90 to 1.04 hours. Successful token use falls from 19.44 to 7.22 million tokens. These changes correspond to reductions of 64.2% and 62.9%, respectively.

Trading performance. CEAU returns 1.53%, compared with 1.44% for direct per-record invocation. The endpoint difference is 0.09 percentage points. CEAU executes two exposure changes instead of three and reduces transaction fees from \$15.03 to \$5.01. Direct per-record invocation records the lower MDD, at 0.27% compared with 0.48%.

The analysis result is clear in this window: CEAU uses 45 fewer triggers and 39 fewer directional assessments. The trading result is mixed. CEAU has a slightly higher return and lower fees, while direct per-record invocation has a lower MDD. The ablation therefore supports CEAU as a control on analysis frequency and aggregate cost. Its trading comparison is limited to this one-week window.

Section 5.6 examines a dense-arrival episode in which both paths reach the same executed position. CEAU uses one analysis, while direct per-record invocation uses six.

5.6 Illustrative Event Traces

Selected event traces illustrate how the evaluated workflows respond to individual evidence arrivals. They explain timing and decision paths; aggregate trading and cost results remain in the preceding tables. The first trace compares EVENT-TRADER with the daily agent baselines. The second compares event-driven formation with fixed schedules. The third compares CEAU with direct per-record invocation during dense evidence arrival.

Intraday evidence and daily agent decisions. Figure 5 presents two long and two short responses from the six-month comparison. The cases cover same-direction, opposite-direction, and no-direction end-of-day (EOD) outcomes among the daily baselines. Their purpose is to show when each method first forms a post-event direction. Aggregate performance is reported in Table 1. The four reports have publication times aligned with their recorded visibility times. The traces also report the return of each EVENT-TRADER position before the daily decision boundary.

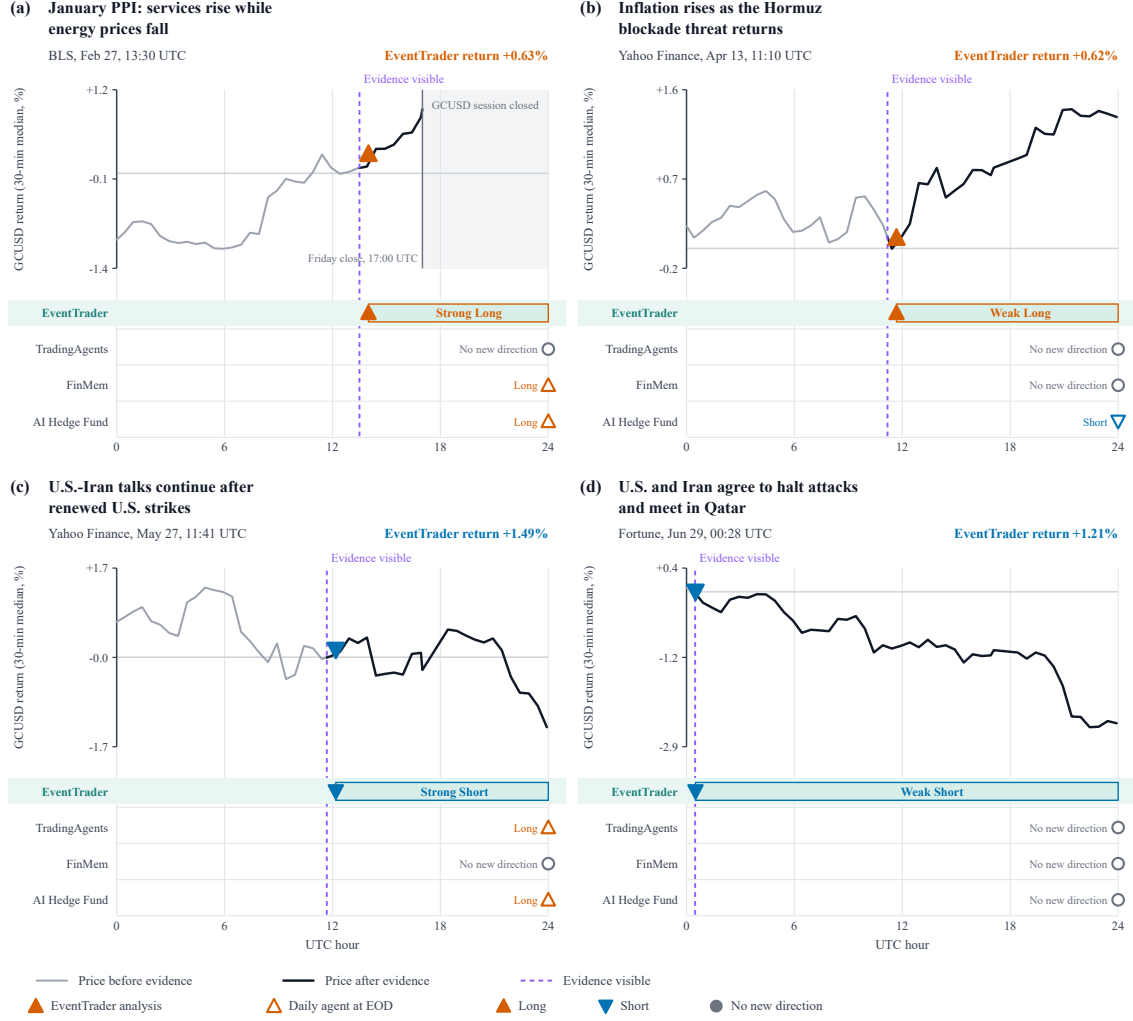


Figure 5: Across four macroeconomic and geopolitical reports, EVENT-TRADER forms an intraday long or short position before the three daily agent baselines act. Each position has a positive return at the final tradable GCUSD bar. No new direction means that no exposure is established after the event boundary. Panel (a) ends at the Friday session close at 17:00 UTC; panels (b)–(d) end at the last bar before the end-of-day (EOD) decision.

Panels (a) and (b) show long responses. The BLS PPI release became visible at 13:30 UTC on February 27. It reported rising service prices and falling energy prices. EVENT-TRADER read the mixed release as leaving the existing bullish setup intact and produced `strong_long` at 14:00 UTC. GCUSD then rose into the Friday close, producing a net return of 0.63%. FinMem and AI Hedge Fund also selected long at the EOD decision, after the session had closed. TradingAgents produced no new direction. In panel (b), Yahoo Finance reported at 11:10:30 UTC that gold was rebounding from a sharp opening decline as inflation and the Hormuz blockade threat returned. EVENT-TRADER produced `weak_long` at 11:40:30 UTC. The position returned 0.62% by EOD. TradingAgents and FinMem produced no new direction, while AI Hedge Fund selected short.

Panels (c) and (d) show short responses. The May 27 Yahoo Finance report described flat gold trading while investors waited for news on U.S.-Iran talks after renewed strikes. EVENT-TRADER placed this narrow range within the existing down leg and produced `strong_short` at 12:11:04 UTC, 30 minutes after the report became visible. GCUSD subsequently fell, producing a net return of 1.49%. TradingAgents and AI Hedge Fund selected long at EOD, while FinMem produced no new direction. On June 29, Fortune reported at 00:28:46 UTC that the United States and Iran had agreed to halt attacks and meet in Qatar. EVENT-TRADER treated the de-escalation as weakening geopolitical support for gold within the active bearish setup. It produced `weak_short` at 00:30 UTC, 1.2 minutes after the report became

visible. GCUSD declined through the day, producing a net return of 1.21%. None of the daily agents established a new direction.

Panel (a) shows that agents can reach the same direction at different decision boundaries. Panels (b)–(d) show EOD decisions that remain flat or take the opposite direction. Across both patterns, EVENT-TRADER forms a position during the event day, while the three baselines defer their decisions to EOD.

Event-driven and fixed schedules. Figure 6 compares two event-driven timing outcomes with the fixed schedules. Panel (a) shows a response before the nearest batch. Panel (b) shows a later response with a more favorable short entry.

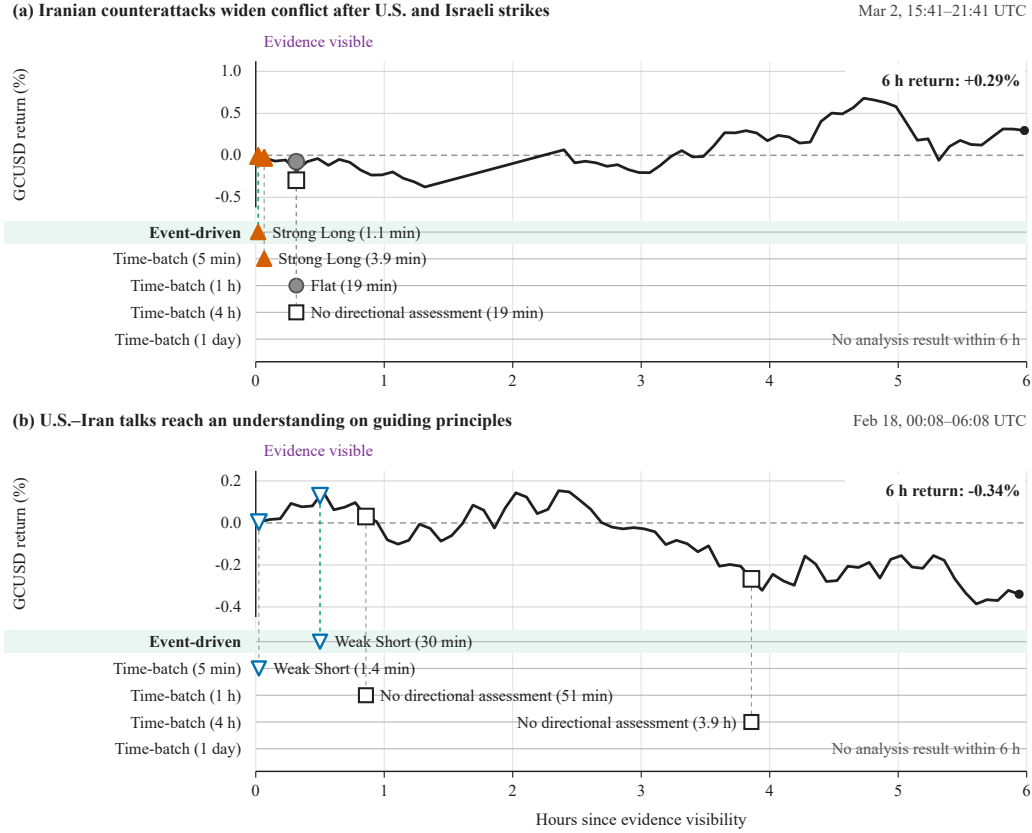


Figure 6: Event-driven and fixed-batch responses to two GCUSD news reports. Panel (a) begins when Yahoo Finance reports at 15:41:03 UTC that Iranian counterattacks have widened the conflict following U.S. and Israeli strikes. Panel (b) begins when FXStreet reports at 00:08:32 UTC that the United States and Iran have reached an understanding on the main guiding principles. Curves show six-hour GCUSD returns after each report becomes visible.

In panel (a), a CBS News price update opened an analysis unit at 15:12:10 UTC. The Yahoo Finance report arrived at 15:41:03 UTC, 1.1 minutes before the unit reached its maturity boundary. It was therefore included in the analysis emitted at 15:42:10 UTC. The analysis treated the wider conflict as support for the safe-haven thesis and the pullback from \$5,400 as consolidation. It produced `strong_long` before the five-minute batch, which reached the same state after 3.9 minutes. GCUSD gained 0.29% over the following six hours.

Panel (b) separates response speed from entry quality. The FXStreet report at 00:08:32 UTC weakened the safe-haven thesis. The five-minute batch produced `weak_short` at 00:09:59 UTC and entered at the next bar open of \$4,950.9. Event-driven kept the analysis unit open until its market-reaction condition was satisfied. It produced `weak_short` at 00:38:32 UTC and entered at \$4,957.6. GCUSD was 0.34% lower at the common six-hour endpoint. For the short position, the \$6.7 higher entry increases the gross return over this horizon from 0.34% to 0.47%. In this case, waiting preserves the short direction and improves its entry price.

Dense evidence arrival. Figure 7 examines the evidence sequence after 12:00 UTC on March 2. Decisions and positions before this boundary are excluded.

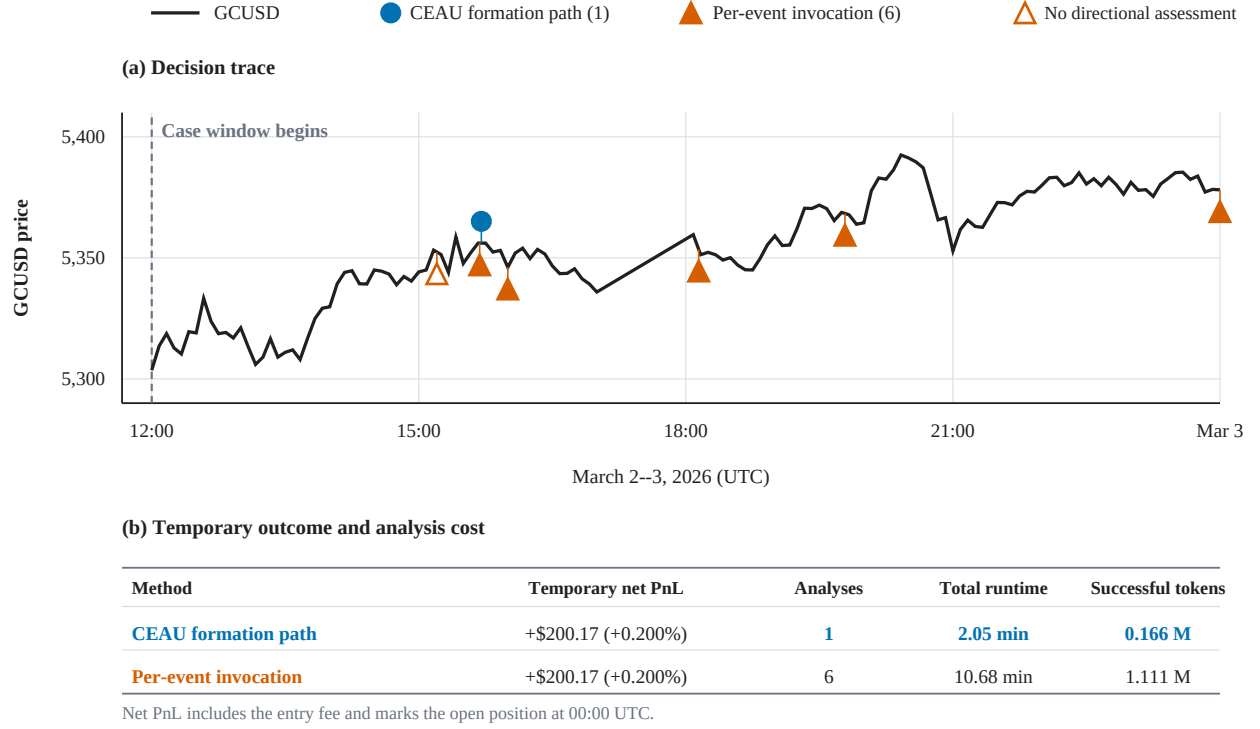


Figure 7: Decision and cost traces during dense evidence arrival on March 2, 2026. Panel (a) shows GCUSD prices and successful analysis outcomes after a 12:00 UTC flat reset. Both paths execute `strong_long` at 15:45 UTC. Panel (b) compares temporary net PnL, total successful analysis runtime, and successful token use through 00:00 UTC on March 3.

Direct per-record invocation runs analysis six times. The first analysis produces no directional assessment, while the next five select `strong_long`. The first directional assessment arrives at 15:41:03 UTC. The CEAU formation path runs analysis once and selects `strong_long` at 15:42:10 UTC. Both assessments therefore take effect at the same 15:45 five-minute bar.

The 1.1-minute difference has no effect on execution. Each path changes exposure once, pays the same \$5.00 entry fee, and produces the same temporary net PnL of \$200.17, or 0.200%, at 00:00 UTC. Their analysis costs differ. The CEAU formation path uses 123 seconds and 0.166 million successful tokens. Direct per-record invocation uses 641 seconds and 1.111 million tokens. This corresponds to reductions of 80.8% in total analysis runtime and 85.1% in successful token use. In this episode, CEAU formation uses five fewer analyses without changing the executed position or its temporary PnL.

6 Related Work

Section 3 uses representative systems to define the time-batched workflow. This section places EVENT-TRADER relative to four broader areas: information in discretionary trading, LLM trading-agent workflows, live and event-centered financial systems, and event-time stream processing.

Information and discretionary research. Financial research starts with a choice about what information deserves attention. Grossman and Stiglitz show that acquiring information has a cost and that informed research must retain economic value Grossman and Stiglitz [1980]. Barber and Odean find that news, unusual trading volume, and extreme returns direct individual investors toward attention-grabbing stocks Barber and Odean [2008]. Tetlock links media pessimism to market activity and short-run price pressure Tetlock [2007]. Shiller studies how widely shared narratives can shape economic behavior Shiller [2017]. These results explain why the arrival and presentation of information affect investment decisions.

Professional investors also differ in how they process information. Hendershott et al. find that institutional order flow contains information about the timing and content of later public news Hendershott et al. [2015]. Akepanidaworn et al. find skill in institutional buying decisions and greater reliance on heuristics when selling Akepanidaworn et al. [2023]. Together, these studies describe discretionary trading as a process of selective attention, interpretation, and portfolio judgment. A new assessment may change the research state even when the current position remains unchanged.

LLM trading-agent workflows. A recent survey describes LLM trading agents as decision pipelines that perceive market information, retrieve context, reason about a decision, produce an action, and adapt through feedback Xia et al. [2026]. Existing systems develop different parts of this pipeline. FinMem combines agent profiling with layered memory and converts retrieved financial information into investment decisions Yu et al. [2023]. TradingGPT uses three memory layers with separate decay rules. It also introduces agent-specific trading traits and inter-agent debate Li et al. [2023].

FinAgent processes numerical data, financial text, and charts. It adds tool use, diversified memory retrieval, and reflection at two levels Zhang et al. [2024]. TradingAgents assigns fundamental, sentiment, and technical analysis to specialized agents. Bull and bear researchers debate their views before a trader and risk team form the final decision Xiao et al. [2024]. FinCon uses a manager-analyst hierarchy. Its risk-control process can start self-critique, update investment beliefs, and send the resulting lessons to selected agents Yu et al. [2024]. TradingGroup adds specialized analysis and forecasting roles, structured self-reflection, dynamic risk controls, and a pipeline that turns trading records into post-training data Tian et al. [2025].

These systems define the memory, reasoning, coordination, and review performed within a trading step. Their workflows begin from observations assembled for that step. Our study examines the preceding runtime question: how newly visible records advance research, how related records form one input, and when that input reaches LLM analysis.

Live and event-centered financial systems. Live and sequential benchmarks make the observations available to trading agents more current. DeepFund uses market data released after model training cutoffs and assigns research, investment decisions, portfolio management, and risk control to separate agents Li et al. [2025]. When Agents Trade provides a lifelong, multi-market environment with verified market data, checked news, and several agent designs with different risk styles Qian et al. [2025]. StockBench evaluates sequential buy, sell, and hold decisions from daily prices, fundamentals, and news Chen et al. [2025a]. LiveTradeBench streams prices and news across stocks and prediction markets. At each trading step, the agent receives the current portfolio and produces percentage allocations Yu et al. [2025]. TiMi separates LLM-based strategy development from minute-level deployment. Its agents develop and refine a trading program that executes at the shorter interval Song et al. [2025]. In each design, an explicit trading step or deployment loop determines when the current state enters decision-making.

Event-centered systems use market events to organize financial information. Janus-Q treats financial news events as primary decision units and links them to observed market responses Li et al. [2026a]. FinKario adds time-sensitive events to a dynamically updated financial knowledge graph and retrieves related graph content for prediction Li et al. [2026b]. These systems use events to structure training data, knowledge, retrieval, or signals. Our study addresses the runtime boundary before LLM analysis: how records about an unfolding market event become visible, form an analysis unit, and advance the research workflow.

Event-time stream processing. Stream-processing systems study when arriving records should produce a higher-level result. SASE filters and correlates records from real-time streams to detect complex event patterns Gyllstrom et al. [2006]. The Dataflow model distinguishes event time from processing time and uses windows, triggers, and watermarks to process unbounded and out-of-order data Akidau et al. [2015]. These works provide the timing concepts used at the stream-to-analysis boundary. We apply this question to LLM financial research: when related evidence records have formed a bounded input that is ready for analysis.

7 Conclusion

LLM trading agents commonly advance through daily or other predefined workflows. These workflows make models, tools, memory, and agent roles available at each decision step. They leave the step itself tied to a schedule. Under asynchronous financial evidence, this creates a timing trade-off between delayed aggregation and repeated analysis. EVENT-TRADER changes what advances the trading-agent loop. Typed messages advance evidence review, analysis, portfolio review, and delayed feedback as their referenced records or artifacts become visible. CEAU then controls when related evidence becomes a bounded input for LLM analysis. The runtime keeps source evidence, revisable research memory, executed portfolio state, and delayed feedback separate. Each operation can therefore be traced to the information visible at its decision time.

Across the six-month GCUSD evaluation, EVENT-TRADER returns 37.68%, the highest return among the evaluated agentic and rule-based methods. In the one-month controlled comparison, the event-driven input-formation path returns 0.47%, while all four fixed-time variants have negative returns. Compared with direct per-record invocation within the event-driven path, CEAU reduces total successful analysis runtime by 64.2% and successful token use by 62.9%. Over the one-week ablation, the two paths return 1.53% and 1.44%, respectively. These results support the evaluated event-driven input-formation path as an alternative to fixed-time workflows. The ablation also shows that CEAU can reduce repeated LLM analysis and aggregate cost within this path.

Limitations. The evidence taxonomy and CEAU formation policy remain configuration-driven. Event labels use a fixed vocabulary, and the present study does not evaluate their classification accuracy. Formation scopes, waiting horizons, and maturity conditions are also set through configuration. The evaluation does not show how these rules transfer across different evidence sources, arrival patterns, markets, or bar resolutions.

The experiments use archived evidence from one asset and MiniMax M2.1. They do not establish the same behavior under multi-asset or other LLM backbones. The runtime supports delayed reflection and controlled memory promotion, but the evaluation does not isolate whether these updates improve later decisions. It also does not attribute the observed trading results to any single agent role.

References

- Yihan Xia, Panpan You, Taotao Wang, Fang Liu, Han Qi, Xiaoxiao Wu, and Shengli Zhang. Agentic trading: When llm agents meet financial markets, 2026.
- Yangyang Yu, Haohang Li, Zhi Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W. Suchow, and Khaldoun Khashanah. Finmem: A performance-enhanced llm trading agent with layered memory and character design, 2023.
- Yang Li, Yangyang Yu, Haohang Li, Zhi Chen, and Khaldoun Khashanah. Tradinggpt: Multi-agent system with layered memory and distinct characters for enhanced financial trading performance, 2023.
- Wentao Zhang, Lingxuan Zhao, Haochong Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, et al. A multimodal foundation agent for financial trading: Tool-augmented, diversified, and generalist, 2024.
- Yijia Xiao, Edward Sun, Di Luo, and Wei Wang. Tradingagents: Multi-agents llm financial trading framework, 2024.
- Yangyang Yu, Zhiyuan Yao, Haohang Li, Zhiyang Deng, Yupeng Cao, Zhi Chen, et al. Fincon: A synthesized llm multi-agent system with conceptual verbal reinforcement for enhanced financial decision making, 2024.
- Changlun Li, Yao Shi, Chen Wang, Qiqi Duan, Runke Ruan, Weijie Huang, Haonan Long, Lijun Huang, Yuyu Luo, and Nan Tang. Time travel is cheating: Going live with deepfund for real-time fund investment benchmarking, 2025.
- Lingfei Qian, Xueqing Peng, Yan Wang, Vincent Jim Zhang, Huan He, Hanley Smith, Yi Han, Yueru He, Haohang Li, Yupeng Cao, Yangyang Yu, Alejandro Lopez-Lira, Peng Lu, Jian-Yun Nie, Guojun Xiong, Jimin Huang, and Sophia Ananiadou. When agents trade: Live multi-market trading benchmark for llm agents, 2025.
- Yanxu Chen, Zijun Yao, Yantao Liu, Jin Ye, Jianing Yu, Lei Hou, and Juanzi Li. Stockbench: Can llm agents trade stocks profitably in real-world markets?, 2025a.
- Zifan Song, Kaitao Song, Guosheng Hu, Ding Qi, Junyao Gao, Xiaohua Wang, Dongsheng Li, and Cairong Zhao. Trade in minutes! rationality-driven agentic system for quantitative financial trading, 2025.
- Xiang Li, Zikai Wei, Yiyan Qi, Wanyun Zhou, Xiang Liu, Penglei Sun, Jian Guo, Yongqi Zhang, and Xiaowen Chu. Janus-q: End-to-end event-driven trading via hierarchical-gated reward modeling, 2026a. URL <https://arxiv.org/abs/2602.19919>.
- Xiang Li, Penglei Sun, Wanyun Zhou, Zikai Wei, Yongqi Zhang, and Xiaowen Chu. Finkario: Event-enhanced automated construction of financial knowledge graph. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pages 9826–9845, 2026b. doi:10.18653/v1/2026.acl-long.446. URL <https://aclanthology.org/2026.acl-long.446/>.
- A. Craig MacKinlay. Event studies in economics and finance. *Journal of Economic Literature*, 35(1):13–39, 1997.
- Sanford J. Grossman and Joseph E. Stiglitz. On the impossibility of informationally efficient markets. *The American Economic Review*, 70(3):393–408, 1980.
- Brad M. Barber and Terrance Odean. All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *The Review of Financial Studies*, 21(2):785–818, 2008.

- Klakow Akepanidaworn, Rick Di Mascio, Alex Imas, and Lawrence Schmidt. Selling fast and buying slow: Heuristics and trading performance of institutional investors. *The Journal of Finance*, 78(6):3055–3098, 2023.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhajan Kambadur, David Rosenberg, and Gideon Mann. Bloomberggpt: A large language model for finance, 2023a.
- Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. Fingpt: Open-source financial large language models, 2023.
- Qianqian Xie, Weiguang Han, Xiao Zhang, Yanzhao Lai, Min Peng, Alejandro Lopez-Lira, and Jimin Huang. Pixiu: A large language model, instruction data and evaluation benchmark for finance, 2023.
- Qianqian Xie, Weiguang Han, Zhengyu Chen, Ruoyu Xiang, Xiao Zhang, et al. Finben: A holistic financial benchmark for large language models, 2024.
- Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, Wayne Xin Zhao, Zhewei Wei, and Ji-Rong Wen. A survey on large language model based autonomous agents, 2023.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.
- Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning, 2023.
- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. Autogen: Enabling next-gen llm applications via multi-agent conversation, 2023b.
- Feng Tian, Flora D. Salim, and Hao Xue. Tradinggroup: A multi-agent trading system with self-reflection and data-synthesis, 2025.
- Jiaxiang Chen, Mingxi Zou, Zhuo Wang, Qifan Wang, Dongning Sun, Chi Zhang, and Zenglin Xu. Finhear: Human expertise and adaptive risk-aware temporal reasoning for financial decision-making, 2025b.
- Haofei Yu, Fenghai Li, and Jiaxuan You. Livetradebench: Seeking real-world alpha with large language models, 2025.
- Tyler Akidau, Robert Bradshaw, Craig Chambers, Slava Chernyak, Rafael J. Fernández-Moctezuma, Reuven Lax, Sam McVeety, Daniel Mills, Frances Perry, Eric Schmidt, and Sam Whittle. The dataflow model: A practical approach to balancing correctness, latency, and cost in massive-scale, unbounded, out-of-order data processing. *Proceedings of the VLDB Endowment*, 8(12):1792–1803, 2015. doi:10.14778/2824032.2824076.
- MiniMax. Minimax m2.1: Significantly enhanced multi-language programming, built for real-world complex tasks. <https://www.minimax.io/news/minimax-m21>, 2025. Released December 23, 2025.
- virattt. AI Hedge Fund. <https://github.com/virattt/ai-hedge-fund>, 2026. Git commit 65a0349ccc114ea936aa08c3a5f1c2b05faa1337.
- Paul C. Tetlock. Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, 62(3):1139–1168, 2007. doi:10.1111/j.1540-6261.2007.01232.x.
- Robert J. Shiller. Narrative economics. *American Economic Review*, 107(4):967–1004, 2017. doi:10.1257/aer.107.4.967.
- Terrence Hendershott, Dmitry Livdan, and Norman Sch"urhoff. Are institutions informed about news? *Journal of Financial Economics*, 117(2):249–287, 2015.
- Daniel Gyllstrom, Eugene Wu, Hee-Jin Chae, Yanlei Diao, Patrick Stahlberg, and Gordon Anderson. Sase: Complex event processing over streams, 2006.